MILITARY SPECIFICATION
CASE, CALIBER .30 CARBINE REFERENCE

1. SCOPE

1.1 This specification covers a primed cartridge case for use in Cartridge, Caliber .30 Carbine Reference.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

C8595418	- Case, Caliber .30 Carbine Reference, Primed
C7643674	- Classification of Cartridge Case Defects
C80538	- Carton, Ball, Carbine Caliber .30 M1
D7553902	- Packing and Marking, Cartridges, Carbine Cal..30; Cartons, Box, Ammunition, M2A1 Wirebound Box
D7553903	- Method of Packing in Box, Ammunition, M2A1
SK FSA 7080	- Method of Packing Cases, Carbine Cal. .30 (Primed or Unprimed) in Cartons, etc.
IEL 8595418	- Inspection Equipment List for Case, Caliber .30 Carbine Reference

PUBLICATIONS

TECP 700-700, Vol. 3 - Manual of Test Methods for Small Arms Ammunition

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(Copies of specifications, standards, drawings and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The primed cartridge case shall comply with all requirements specified on drawing C8595418 and the following:

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Function and casualty.- The primed cartridge case, when assembled into a cartridge shall function without casualty.

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The cartridge case shall be free of wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.5.2 Foreign matter.- The cartridge case shall be free of corrosion, stains, discoloration, dirt, oil or smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 200 cartridge cases. The samples shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply

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4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of cartridge cases shall be identified as to type, caliber and model, as well as with a lot number and the manufacturer's identification as assigned by the procuring activity.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 1.00 percent and the AQL for the minor class shall be 2.50 percent. All non-conforming cartridge cases shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and Method of Inspection	Major	Minor	Major or Minor
	Visual 1/			
1	Discolored, dirty, oily, smeared		X	
2	Corroded or stained (if etched)	X		
3	Mixed Cases	X		
4	Round head	X		
5	Dent			X
6	Split case	X		
7	Perforated case	X		
8	Draw scratch			X
9	Scratch		X	
10	Beveled underside of head	X		
11	Scaly metal			X
12	No chamfer on head (rim)	X		

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4.3.2.1 (Cont'd)

TABLE I (Cont'd)

No.	Defect and Method of Inspection	Major	Minor	Major or Minor
	Visual 1/ (Cont'd)			
13	Wrinkle		X	
14	Bulge		X	
15	Illegible or missing head stamp		X	
16	Defective head		X	
17	Defective mouth		X	
26	Primer missing	X		
27	Cocked primer	X		
28	Inverted primer	X		
29	Loose primer	X		
30	Nicked or dented primer		X	
31	No waterproofing material (Primer pocket joint)		X	
32	Primer Vent missing	X		
	Gaging			
33	Total length	X		
34	Case profile failure	X		
35	Diameter of mouth	X		
36	Diameter of extractor groove, max.	X		
37	Diameter of extractor groove, min.		X	
38	Diameter of head	X		
39	Thickness of head	X		
40	Primer depth	X		
	Weighing			
41	Weight	X		

1/ Refer to Standard MIL-STD-636 for visual standards for defects 1 through 31. Should a conflict exist between Standard MIL-STD-636 and the classification of defects in Table I, the classification of such defect in Table I shall prevail.

4.3.2.2 Packaging, Packing and Marking Inspection.- Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644 as applicable.

4.3.3 Tests.- The tests shall be conducted in accordance with the methods and procedures specified in 4.4, except as provided in 4.3.

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4.3.3.1 Residual stress (Mercurous nitrate).- A sample of 20 primed cases shall be selected from the lot for the mercurous nitrate test. Failure of two or more cartridge cases to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge case fails in the initial test, a second sample consisting of double the number of cartridge cases in the first sample may be tested. If any failing cartridge cases are found in the second sample, the lot shall be rejected.

4.3.3.2 Function and casualty.- Fifty primed cases shall be selected from the lot and tested in accordance with 4.4.2. The lot shall be rejected when function and casualty defects exceed the acceptance number for the cumulative sample in Table II. If the number of defects found in the initial test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantity of cartridge cases in the first sample, may be tested. If the total number of defects in the combined first and second sample, exceeds the acceptance number for the cumulative sample, the lot shall be rejected.

TABLE II

Defect	Acceptance Number	
	1st Sample	Cumulative (1st & 2nd Sample
1. Misfire	0	1
2. Primer leak		
a. Perforation in firing pin indent in primer cup	0	2
b. Escape of gas through primer cup other than a.	0	2
c. Escape of gas around primer cup		
(1) More than 50% of periphery	1	3
(2) Less than 50% of periphery	2	6
d. Loose primer		
(1) Blown primer or primer falls out of pocket on retraction of bolt	0	1
(2) Primer remains in pocket but is physically loose	1	3
3. Case casualties		
a. Longitudinal split 1/		
(1) Neck and shoulder (I and S)	2	6
(2) Body (J)	1	3
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 1/		
(1) Partial, shoulder or body (S or J)	0	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
4. Failure to extract	0	1

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1/ For location of defects indicated by letters in parenthesis, see drawing C7643674.

4.3.4 Inspection Equipment.- The examination and tests shall be made using equipment listed on IEL-8595418 except as provided in 4.3.

4.4 Test methods and procedures.

4.4.1 Residual stress (mercurous nitrate).- The test shall be conducted in accordance with TECP 700-700, Vol. 3.

4.4.2 Function and casualty.- The cartridge cases for this test shall be loaded with sufficient propellant to produce an average velocity of 1900 feet per second (ft/sec) plus or minus 30 ft/sec at 53 feet from the muzzle when assembled into reference cartridges and fired in the appropriate test equipment. Each case, upon removal after firing, shall be examined and defects recorded.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Primed cartridge cases shall be packaged in cartons, Drawing C80538 in accordance with method shown on drawing SK FSA 7080. The cartons shall be packed in M2A1 ammunition boxes in accordance with drawing D7553903.

5.2 Packing.- The M2A1 boxes shall be packed in a wirebound crate in accordance with drawing D7553902.

5.3 Marking.- Each carton, ammunition box and wirebound crate shall be marked on the outside surface with the primed case nomenclature, quantity, manufacturer's identification and assigned lot number.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of cartridge cases presented to the Government.

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6.1.4 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1305 - A482

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